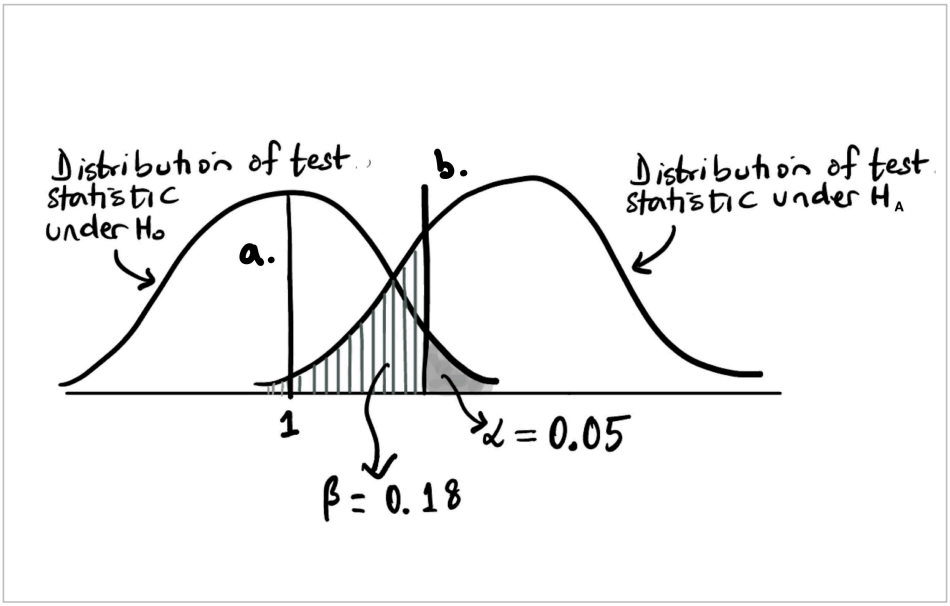
Warmup/Attendance: [Week 10: Friday 10/30/2026]

Full Name: SID:

Work with the people around you to try to solve these problems, which we'll discuss together once lecture begins. These are not graded for correctness, but you will turn this sheet in for attendance at the end of class.

**1.** The following two questions are based on the diagram below. The diagram corresponds to a one-sided hypothesis test that uses the following null hypothesis: $H_0 : \mu = 1$, and a significance level of $\alpha = 0.05$. Assume that the distribution on the left corresponds to the distribution of the test statistic under $H_0$ and that the distribution on the right represents a *possible* distribution under $H_A$.




- What is an alternative hypothesis that corresponds to the distribution under $H_A$ you see?

- How does a Type I error occur (under which distribution and on what side of line **b.**)?

- How does a Type II error occur (under which distribution and on what side of line **b.**)?

- Calculate the power of this test.